

§ 44. General proof of the principal theorem.

We now suppose that the fundamental theorem has been proved for symmetric functions of $n - 1$ variables, and we derive it, by a procedure entirely analogous to that applied in § 43, for n variables.

Let again

$$S = S(\alpha_1, \alpha_2, \dots, \alpha_n) \quad (1)$$

be a whole symmetric function of the n variables $\alpha_1, \alpha_2, \dots, \alpha_n$. If we arrange it according to powers of α_1 , and accordingly put

$$S = S_0\alpha_1^\mu + S_1\alpha_1^{\mu-1} + \dots + S_{\mu-1}\alpha_1 + S_\mu, \quad (2)$$

then the coefficients $S_0, S_1, \dots, S_\mu$ are whole symmetric functions of the $n - 1$ variables $\alpha_2, \dots, \alpha_n$.

If we denote the symmetric fundamental functions of these latter variables by

$$a'_1, a'_2, \dots, a'_{n-1},$$

then, by our supposition, the coefficients $S_0, S_1, \dots, S_\mu$ can be expressed rationally by these. But by § 42, (2) and (3),

$$\begin{aligned} a'_1 &= f_1(\alpha_1) = \alpha_1 + a_1, \\ a'_2 &= f_2(\alpha_1) = \alpha_1^2 + a_1\alpha_1 + a_2, \\ a'_3 &= f_3(\alpha_1) = \alpha_1^3 + a_1\alpha_1^2 + a_2\alpha_1 + a_3, \end{aligned}$$

and so on. Thus the coefficients $S_0, S_1, \dots, S_\mu$ can be expressed wholly and rationally by

$$\alpha_1, a_1, a_2, \dots, a_n.$$

If, therefore, after these expressions have been introduced, we arrange (2) anew according to powers of α_1 , we obtain

$$S = A_0\alpha_1^m + A_1\alpha_1^{m-1} + \dots + A_{m-1}\alpha_1 + A_m, \quad (3)$$

where in general m is an exponent different from μ , and where the coefficients $A_0, A_1, \dots, A_m$ depend wholly and rationally on $a_1, a_2, \dots, a_n$. We put again

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \dots + A_{m-1}x + A_m \quad (4)$$

and divide $\Phi(x)$ by

$$\begin{aligned} f(x) &= x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n \\ &= (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n). \end{aligned}$$

There results a quotient and a remainder which, with respect to x , is at most of degree $n - 1$. We therefore put

$$\Phi(x) = Qf(x) + \psi(x), \quad (5)$$

$$\psi(x) = C_0x^{n-1} + C_1x^{n-2} + \dots + C_{n-2}x + C_{n-1}, \quad (6)$$

and here $C_0, C_1, \dots, C_{n-1}$ are whole rational functions of $a_1, a_2, \dots, a_n$, in which, if S in its original form contains no fractional coefficients, no fractions occur either.

But now, since $f(\alpha_1)$ vanishes and $S = \Phi(\alpha_1)$, it follows from (5) that

$$S = \psi(\alpha_1). \quad (7)$$

Since S is symmetric, α_1 may here be replaced by $\alpha_2, \alpha_3, \dots, \alpha_n$. This gives the following n equations:

$$\begin{aligned} C_0\alpha_1^{n-1} + C_1\alpha_1^{n-2} + \dots + C_{n-2}\alpha_1 + (C_{n-1} - S) &= 0, \\ C_0\alpha_2^{n-1} + C_1\alpha_2^{n-2} + \dots + C_{n-2}\alpha_2 + (C_{n-1} - S) &= 0, \\ &\vdots \\ C_0\alpha_n^{n-1} + C_1\alpha_n^{n-2} + \dots + C_{n-2}\alpha_n + (C_{n-1} - S) &= 0. \end{aligned} \tag{8}$$

If we regard this system as a system of homogeneous linear equations with the n unknowns

$$C_0, C_1, \dots, C_{n-2}, C_{n-1} - S,$$

then its determinant is

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \dots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \dots & \alpha_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \dots & \alpha_n & 1 \end{vmatrix},$$

which, by § 22, formula (12), is equal to the product of all differences

$$\alpha_1 - \alpha_2, \quad \alpha_1 - \alpha_3, \dots$$

and is therefore different from zero. Consequently, by Theorem II in § 24,

$$C_0 = 0, \quad C_1 = 0, \dots, \quad C_{n-2} = 0,$$

and

$$S = C_{n-1}, \tag{9}$$

which contains the fundamental theorem to be proved.

One reaches the same result also by Theorem II, § 30; for since the function of degree $n - 1$ in x ,

$$C_0x^{n-1} + C_1x^{n-2} + \dots + C_{n-2}x + C_{n-1} - S,$$

vanishes for $x = \alpha_1, \alpha_2, \dots, \alpha_n$, that is, for more than $n - 1$ values, all its coefficients must be zero.

The proof which we have here given for the fundamental theorem in connection with Cauchy¹ at the same time offers a means, in special cases, actually to compute the expression of a symmetric function by the fundamental functions. The other proofs known for the theorem offer the same possibility. We shall communicate here a second proof, which leads to a method of computation that is often simpler.²

§ 45. Second proof of the theorem on symmetric functions.

Let S be a whole symmetric function of the variables $\alpha_1, \alpha_2, \dots, \alpha_n$. The individual terms of this function are all of the form

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\dots\alpha_n^{\nu_n},$$

where M is a coefficient independent of the α 's. These terms shall now be ordered in a definite manner. After the order of the variables $\alpha_1, \alpha_2, \dots, \alpha_n$ has been fixed, of two terms

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\dots\alpha_n^{\nu_n}, \quad A' = M'\alpha_1^{\nu'_1}\alpha_2^{\nu'_2}\dots\alpha_n^{\nu'_n},$$

¹Cauchy, *Exercices de mathématiques*, fourth year.

²Waring, *Meditationes algebraicae*. Gauss, *Demonstratio nova altera theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse*. Werke, vol. III, p. 36.

A shall be called the higher if the first of the differences

$$\nu_1 - \nu'_1, \quad \nu_2 - \nu'_2, \dots, \nu_n - \nu'_n$$

which is different from zero has a positive value; thus if either $\nu_1 > \nu'_1$, or $\nu_1 = \nu'_1, \nu_2 > \nu'_2$, or $\nu_1 = \nu'_1, \nu_2 = \nu'_2, \nu_3 > \nu'_3$, and so on.

Since we suppose that all terms in which all exponents $\nu_1, \nu_2, \dots, \nu_n$ agree have been united into one term, it is thereby decided, for any two terms, which is the higher; and if A is higher than A' , and A' higher than A'' , then A is also higher than A'' .

Let now, according to this ordering,

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}$$

be the highest term of our function S . It follows that

$$\nu_1 \geq \nu_2$$

must hold; for if $\nu_1 < \nu_2$, then the term

$$M\alpha_1^{\nu_2}\alpha_2^{\nu_1}\cdots\alpha_n^{\nu_n},$$

which must also occur in S because of symmetry, would stand higher in the order than A , contrary to the supposition. In the same way it follows that $\nu_2 \geq \nu_3$ must hold; for if $\nu_2 < \nu_3$, then

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_3}\alpha_3^{\nu_2}\cdots\alpha_n^{\nu_n}$$

would stand higher than A , and so forth. We therefore conclude that the exponents in A ,

$$\nu_1, \nu_2, \nu_3, \dots, \nu_n,$$

form a decreasing, or at least never increasing, sequence of numbers; that is, the differences

$$\nu_1 - \nu_2, \quad \nu_2 - \nu_3, \dots, \nu_{n-1} - \nu_n$$

are all positive, or at least not negative.

We conclude secondly that in no term of the function S can an exponent higher than ν_1 occur; for otherwise there would also occur a term in which α_1 had this higher exponent, and that would be of higher order than A , contrary to the supposition. Thus the terms that can occur at all in a symmetric function are, apart from the coefficients, completely determined by the highest term; and the number of possible terms is finite when the highest term is given.

We can denote the order of the symmetric function by the exponent sequence of its highest term,

$$(\nu_1, \nu_2, \dots, \nu_n).$$

Apart from the case in which all exponents are zero, so that the function is a constant, the lowest possible order is

$$(1, 0, \dots, 0),$$

to which corresponds the single symmetric function

$$M(\alpha_1 + \alpha_2 + \cdots + \alpha_n) = -Ma_1.$$

The next higher order is

$$(1, 1, 0, \dots, 0),$$

to which corresponds the symmetric function

$$Ma_1 + M'a_2,$$

and so in the first n orders we obtain the fundamental functions themselves and their linear combinations, and no others.

The next following order is characterized by

$$(2, 0, 0, \dots, 0),$$

and contains the linear combinations of the fundamental functions with the sum of the squares.

In all these cases the representability of the symmetric functions has already been proved; and we shall now carry out the general proof by deriving the representation of the function S , of order $(\nu_1, \nu_2, \dots, \nu_n)$, by the fundamental functions under the supposition that it is already known for functions of lower order, thus by the procedure of complete induction.

We note for this purpose that, by multiplying two symmetric functions S and S' , whose highest terms are

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}, \quad A' = M'\alpha_1^{\nu'_1}\alpha_2^{\nu'_2}\cdots\alpha_n^{\nu'_n},$$

there arises a symmetric function whose highest term is the product

$$AA' = MM'\alpha_1^{\nu_1+\nu'_1}\alpha_2^{\nu_2+\nu'_2}\cdots\alpha_n^{\nu_n+\nu'_n}.$$

For suppose that in SS' there were a higher term than AA' , arising from the multiplication of the two terms

$$B = N\alpha_1^{\mu_1}\alpha_2^{\mu_2}\cdots\alpha_n^{\mu_n}, \quad B' = N'\alpha_1^{\mu'_1}\alpha_2^{\mu'_2}\cdots\alpha_n^{\mu'_n},$$

so that

$$BB' = NN'\alpha_1^{\mu_1+\mu'_1}\alpha_2^{\mu_2+\mu'_2}\cdots\alpha_n^{\mu_n+\mu'_n},$$

and at the same time not $B = A$ and $B' = A'$. Then the first of the differences

$$\mu_1 - \nu_1 + \mu'_1 - \nu'_1, \quad \mu_2 - \nu_2 + \mu'_2 - \nu'_2, \dots, \quad \mu_n - \nu_n + \mu'_n - \nu'_n$$

which is not zero would be positive; this is impossible, since the first non-vanishing one among the differences

$$\mu_1 - \nu_1, \quad \mu_2 - \nu_2, \dots, \mu_n - \nu_n$$

and among the differences

$$\mu'_1 - \nu'_1, \quad \mu'_2 - \nu'_2, \dots, \mu'_n - \nu'_n$$

is, by supposition, negative. By repeating this conclusion one obtains the more general theorem that the highest term of a product of several factors is obtained by multiplying the highest terms of the individual factors.

The highest terms of the symmetric fundamental functions

$$a_1, a_2, a_3, \dots, a_n$$

are now, apart from the sign,

$$\alpha_1, \quad \alpha_1\alpha_2, \quad \alpha_1\alpha_2\alpha_3, \dots, \alpha_1\alpha_2\alpha_3\cdots\alpha_n;$$

and if therefore we form the product

$$P = \pm M a_1^{\nu_1-\nu_2} a_2^{\nu_2-\nu_3} \cdots a_{n-1}^{\nu_{n-1}-\nu_n} a_n^{\nu_n},$$

we obtain a symmetric function whose highest term is likewise A , as in S .

The difference

$$S - P = S'$$

is therefore again a symmetric function whose highest term is lower than that of S , and which, by the supposition, we can represent rationally by the fundamental functions. Since P is represented in

the same way, the goal is reached. Here again it is clear that through the representation by means of the fundamental functions no fractions are introduced if originally none are contained in S .

This method of computing symmetric functions at the same time gives information about the degree of the whole rational function of $a_1, a_2, \dots, a_n$ by which a definite symmetric function of the $\alpha_1, \alpha_2, \dots, \alpha_n$ is expressed. Namely, if $(\nu_1, \nu_2, \dots, \nu_n)$ is the order of the symmetric function, measured by the exponent sequence of the first term, then ν_1 is the degree of P with respect to all $a_1, a_2, \dots, a_n$, and in the developed expression for S no term of higher degree can occur. Nor can the term P be destroyed by any of the following terms, since P is completely determined by the order of S , and the order of S' , and of all following symmetric functions, is lower than the order of S .

If therefore, in place of $a_1, a_2, \dots, a_n$, we put

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \dots, \frac{a_n}{a_0},$$

then $a_0^{\nu_1} S$ is a whole homogeneous function of degree ν_1 of $a_0, a_1, \dots, a_n$, and one cannot transform S into a whole function of the a 's by multiplication with a lower power of a_0 .

§ 46. Discriminants.

The theorems obtained above teach how to express symmetric functions of the roots of an equation rationally by the coefficients of the equation. The results were first derived under the supposition that the roots, or, what by the third section is equivalent to this, the coefficients were independent variables; but they cannot become false when definite numerical values are substituted for them, since at least for whole symmetric functions no denominators occur which could vanish.

We first make some applications to special symmetric functions which are important for what follows. Already in § 19 it was remarked that the product of differences

$$P = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n) \cdots (\alpha_{n-1} - \alpha_n) \quad (1)$$

changes only its sign when two elements are interchanged. The square of this product will therefore remain unchanged under all interchanges of two of the variables α , and hence also under all permutations; that is, it is a symmetric function of the variables. The order of this symmetric function is, as one sees from (1),

$$(2n - 2, 2n - 4, \dots, 2, 0).$$

If therefore we take the function $f(x)$, whose roots are $\alpha_1, \alpha_2, \dots, \alpha_n$, in the form

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n,$$

so that the highest power of x has not necessarily the coefficient 1, then P^2 is a whole rational function of

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \dots, \frac{a_n}{a_0}, \quad (2)$$

which, by multiplication by a_0^{2n-2} , passes into a homogeneous function of $a_0, a_1, \dots, a_n$. We call this homogeneous function, which is therefore defined by

$$a_0^{2n-2} P^2 = D, \quad (3)$$

the discriminant of the function $f(x)$. If we put $a_0 = 1$, we obtain the non-homogeneous form of the discriminant.³

³The discriminant is, apart from the factor $(-1)^{\frac{n(n-1)}{2}} a_0^{2n-2}$, the same as what Gauss calls the determinant of the function (loc. cit., art. 6).

The vanishing of the discriminant is the necessary and sufficient condition that two of the values $\alpha_1, \alpha_2, \dots, \alpha_n$ become equal.

We can set up a second expression for the discriminant. By § 29, (4),

$$\begin{aligned} f'(\alpha_1) &= a_0(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n), \\ f'(\alpha_2) &= a_0(\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n), \\ &\vdots \\ f'(\alpha_n) &= a_0(\alpha_n - \alpha_1)(\alpha_n - \alpha_2) \cdots (\alpha_n - \alpha_{n-1}), \end{aligned}$$

and consequently, since here every factor $(\alpha_h - \alpha_k)$ occurs twice with opposite signs,

$$D = (-1)^{\frac{n(n-1)}{2}} a_0^{n-2} f'(\alpha_1) f'(\alpha_2) \cdots f'(\alpha_n). \quad (5)$$

For actually representing the discriminant we first have a general means. By § 22,

$$(-1)^{\frac{n(n-1)}{2}} P = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{n-1} \end{vmatrix}, \quad (6)$$

and if, by § 27, we represent the square of this determinant by multiplication according to vertical rows as a new determinant, and put

$$s_m = \alpha_1^m + \alpha_2^m + \cdots + \alpha_n^m,$$

then

$$D = a_0^{2n-2} \begin{vmatrix} s_0 & s_1 & s_2 & \cdots & s_{n-1} \\ s_1 & s_2 & s_3 & \cdots & s_n \\ \vdots & \vdots & \vdots & & \vdots \\ s_{n-1} & s_n & s_{n+1} & \cdots & s_{2n-2} \end{vmatrix}. \quad (7)$$

The quantities s_m , however, are the power sums which we have already expressed by the coefficients in § 42; only in place of $a_1, a_2, \dots, a_n$ the quotients (2) are to be put.

Thus we find, for example, for $n = 2$

$$D = a_0^2(s_0 s_2 - s_1^2) = a_1^2 - 4a_0 a_2, \quad (8)$$

and for $n = 3$

$$D = a_0^4(s_0 s_2 s_4 - s_0 s_3^2 + 2s_1 s_2 s_3 - s_1^2 s_4 - s_2^3). \quad (9)$$